ĐẠI HỌC QUỐC GIA HÀ NỘI
TRƯỜNG ĐẠI HỌC CÔNG NGHỆ

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KHOÁ LUẬN TỐT NGHIỆP

Ngành: Công nghệ thông tin

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RESEARCH ON INTRINSIC MOTIVATION FOR REINFORCEMENT LEARNING

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Declaration of Authorship

I hereby declare that this thesis is my own work and has not been submitted previously for a degree or diploma at this or any other higher education institution. To the best of my knowledge and belief, the thesis contains no material previously published or written by another person except where proper acknowledgment and citation are given.

Hanoi, ...

Student

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Tóm tắt

Adaptive Correlation-Weighted Intrinsic Rewards for Reinforcement Learning nghiên cứu bài toán học tăng cường trong môi trường phần thưởng thưa, nơi tác tử rất khó khám phá do tín hiệu phản hồi từ môi trường xuất hiện không thường xuyên. Nhiều phương pháp hiện nay kết hợp phần thưởng ngoại sinh với phần thưởng nội tại dựa trên tò mò hoặc độ mới, nhưng thường phải dùng một hệ số cố định để cân bằng hai nguồn phần thưởng này. Hệ số cố định dễ nhạy với từng bài toán và không phản ánh được việc ở những trạng thái khác nhau thì nhu cầu khám phá cũng khác nhau.

Khóa luận trình bày ACWI, một cơ chế điều chỉnh phần thưởng nội tại theo từng trạng thái thông qua một Beta Network gọn nhẹ. Mạng này được huấn luyện bằng mục tiêu tương quan nhằm làm cho phần thưởng nội tại sau khi được scale có quan hệ chặt chẽ hơn với tổng phần thưởng ngoại sinh trong tương lai. Phương pháp được kết hợp với PPO và Intrinsic Curiosity Module để vừa tăng cường khả năng khám phá, vừa giữ được tính ổn định trong quá trình huấn luyện.

Kết quả thực nghiệm trên năm môi trường MiniGrid cho thấy ACWI cải thiện hiệu quả mẫu và giảm độ dao động so với PPO thuần túy cũng như PPO kết hợp ICM với hệ số cố định. Phương pháp đặc biệt hiệu quả trong các môi trường có phần thưởng thưa nhưng vẫn còn tín hiệu ngoại sinh mang tính định hướng. Trong những bài toán quá thưa, ACWI suy giảm một cách ổn định về gần hành vi của hệ số cố định thay vì gây mất ổn định cho quá trình học.

Từ khóa: học tăng cường, phần thưởng thưa, phần thưởng nội tại, khám phá dựa trên tò mò, điều chỉnh phần thưởng thích ứng

Abstract

Sparse-reward reinforcement learning remains difficult because agents receive limited feedback about whether their behavior is useful. A common solution is to augment the environment reward with intrinsic motivation signals such as novelty, curiosity, or impact. However, count-based bonuses, the Intrinsic Curiosity Module (ICM), and Rewarding Impact-Driven Exploration (RIDE) are all commonly combined with extrinsic rewards through a fixed coefficient. This design makes performance highly sensitive to manual tuning and obscures a broader empirical question: how does an adaptive scaling mechanism affect different intrinsic reward modules?

This thesis frames *Adaptive Correlation-Weighted Intrinsic Rewards for Reinforcement Learning* as a module-agnostic adaptive wrapper. ACWI learns a state-dependent scaling factor through a lightweight Beta Network and uses a correlation-based objective to align weighted intrinsic rewards with discounted future extrinsic returns. Under this framing, ACWI is not treated only as a standalone method paired with one curiosity bonus, but as a mechanism for studying and improving how representative intrinsic reward modules are used during training.

The thesis therefore analyzes ACWI with respect to three representative intrinsic reward families: count-based exploration, ICM, and RIDE. In the implemented system, the quantitative experiments instantiate the framework with Proximal Policy Optimization and ICM on five sparse-reward MiniGrid environments, while count-based and RIDE modules are used to motivate the comparative experimental framing and discussion. The results show that adaptive scaling improves sample efficiency and reduces variance relative to PPO without intrinsic rewards and PPO with fixed intrinsic coefficients. Overall, the study suggests that ACWI is best understood as a practical adaptive layer whose benefits depend on the structure and informativeness of the underlying intrinsic reward module.

Keywords: reinforcement learning, sparse rewards, intrinsic motivation, adaptive reward scaling, count-based exploration, ICM, RIDE

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List of Acronyms

Acronym	**Full Form**	**Description**
ACWI	Adaptive Correlation-Weighted Intrinsic	Proposed adaptive intrinsic reward scaling framework
RL	Reinforcement Learning	Learning paradigm based on sequential interaction with an environment
PPO	Proximal Policy Optimization	On-policy policy-gradient algorithm used as the training backbone
ICM	Intrinsic Curiosity Module	Intrinsic reward model based on forward prediction error
RND	Random Network Distillation	Curiosity-based method using predictor error against a random target network
RIDE	Rewarding Impact-Driven Exploration	Intrinsic reward method based on state-change impact
GAE	Generalized Advantage Estimation	Technique for reducing variance in policy-gradient advantage estimates
MDP	Markov Decision Process	Standard formal model for reinforcement learning environments

Chapter 1

Introduction

1.1 Background and Motivation

Reinforcement learning (RL) has become a central paradigm for training agents to solve sequential decision-making problems, with influential successes in Atari games, Go, and robotic control [1–3]. In many of these achievements, the learning process benefits from dense reward signals that provide frequent guidance about which behaviors are desirable. However, many practical environments are sparse-reward: the agent receives useful feedback only after a long sequence of actions, making exploration substantially harder [4].

To mitigate this problem, a large body of research augments the extrinsic reward supplied by the environment with an intrinsic reward that encourages exploration [5, 6]. Count-based methods reward the agent for visiting rarely explored states [7–9]. Curiosity-driven methods instead derive intrinsic rewards from prediction errors or representation discrepancies, as in the Intrinsic Curiosity Module (ICM) [10], Random Network Distillation (RND) [11], and Rewarding Impact-Driven Exploration (RIDE) [12]. These methods often improve exploration, but they still rely on a crucial design choice: how strongly intrinsic rewards should be weighted relative to extrinsic rewards.

In many existing approaches, the balance between the two reward sources is controlled by a fixed scalar coefficient. This choice is usually tuned manually for each environment, and the best value can vary significantly across tasks and training stages [13, 14]. A constant coefficient cannot distinguish between states where continued exploration is useful and states where exploration has become distracting. More importantly, the same weakness appears across multiple intrinsic reward families, from count-based bonuses to ICM and RIDE. This observation motivates a broader thesis framing: instead of studying ACWI only as a new method attached to one module, we study it as an adaptive mechanism whose empirical impact can be analyzed across representative intrinsic reward modules.

1.2 Research Problem

This thesis investigates the following problem: how does adaptive intrinsic reward scaling affect different intrinsic reward modules, and can such scaling promote useful exploration in sparse-reward reinforcement learning without destabilizing policy optimization?

More specifically, the thesis focuses on state-dependent reward modulation. Let the agent receive an extrinsic reward R_t^E from the environment and an intrinsic reward $R_t^{I,m}$ produced by intrinsic reward module m, where m may denote a count-based bonus, ICM, or RIDE. Standard methods optimize a shaped reward of the form

$$\bar{r}_t = R_t^E + \beta R_t^{I,m}, \tag{1.1}$$

where β is a fixed hyperparameter. The limitation of this formulation is that the same intrinsic scaling is applied uniformly to all states, even though the value of exploration may differ substantially across the state space.

The core research question of this thesis is therefore:

> Can ACWI serve as a module-agnostic, state-dependent scaling mechanism that improves how representative intrinsic reward modules such as count-based bonuses, ICM, and RIDE contribute to learning?

1.3 Research Objectives and Scope

The objective of the thesis is to adapt the paper *Adaptive Correlation-Weighted Intrinsic Rewards for Reinforcement Learning* into a thesis-style empirical study of how ACWI changes the behavior of intrinsic reward modules. The specific objectives are as follows:

- analyze why fixed intrinsic reward coefficients are a shared limitation across representative intrinsic reward modules;
- review count-based exploration, ICM, RIDE, and prior adaptive reward weighting approaches;
- present ACWI as a lightweight, state-dependent scaling mechanism that can wrap different intrinsic reward generators;

- describe the concrete implementation of ACWI with PPO and ICM while clarifying how the same formulation extends to count-based and RIDE modules;
- evaluate the empirical effect of adaptive scaling in sparse-reward MiniGrid environments and discuss what the results imply for different intrinsic reward modules.

The scope of the thesis is limited to episodic RL tasks with sparse rewards. Conceptually, the study is framed around three representative intrinsic reward families: count-based bonuses, ICM, and RIDE. In implementation, however, the quantitative experiments use PPO as the base policy optimization algorithm and ICM as the instantiated intrinsic reward generator. Count-based and RIDE modules are included in the literature review and experimental framing to position ACWI as a general adaptive layer rather than an ICM-specific technique. The experiments focus on benchmark MiniGrid environments rather than real-world robotics or continuous-control tasks. Likewise, this thesis emphasizes empirical performance and methodological clarity instead of a full theoretical proof for convergence or optimality.

1.4 Research Methodology

The work follows a research-oriented methodology. First, the thesis surveys foundational RL literature and representative exploration modules to identify the common limitation of fixed intrinsic weighting. Second, it formulates ACWI as an adaptive scaling mechanism that predicts a positive coefficient $\beta(s_t)$ from the current state and can be applied to module-specific intrinsic rewards. Third, the method is instantiated in a PPO training loop using ICM-derived intrinsic rewards as the primary quantitative case study. Finally, the approach is validated experimentally on multiple sparse-reward tasks in MiniGrid by comparing it against PPO without intrinsic rewards and PPO with fixed intrinsic coefficients, and by interpreting the findings through the broader lens of count-based, ICM, and RIDE-style intrinsic signals.

The evaluation criteria emphasize sample efficiency, learning stability, robustness across environments, and the qualitative behavior of the learned adaptive scaling policy. This combination of theoretical grounding, algorithmic design, and empirical validation is consistent with the structure of a machine learning graduation thesis and supports an experimental framing centered on ACWI's effect on intrinsic reward modules.

1.5 Thesis Structure

The remainder of this thesis is organized as follows. Chapter 2 reviews reinforcement learning with intrinsic motivation, representative intrinsic reward modules, and prior attempts to adapt the weighting of intrinsic rewards. Chapter 3 presents the proposed methodology, including the module-agnostic ACWI formulation, the concrete ICM instantiation, and the training algorithm. Chapter 4 describes the experimental setup and analyzes the results on MiniGrid benchmark environments from the perspective of ACWI's effect on intrinsic reward modules. Finally, the conclusion summarizes the main findings, discusses limitations, and outlines future research directions.

Chapter 2

Literature Review

2.1 Reinforcement Learning in Sparse-Reward Environments

An episodic reinforcement learning problem is commonly modeled as a Markov Decision Process (MDP) defined by the tuple $(\mathcal{S}, \mathcal{A}, P, R, \gamma, \rho_0)$, where $\mathcal{S}$ is the state space, $\mathcal{A}$ is the action space, P denotes the transition dynamics, R is the reward function, $\gamma \in [0, 1)$ is the discount factor, and ρ_0 is the initial state distribution [3]. The goal is to learn a policy $\pi_\theta(a \mid s)$ that maximizes the expected discounted return:

$$J(\theta) = \mathbb{E}_{\tau \sim (\pi_\theta, P)} \left[\sum_{t=0}^{T-1} \gamma^t R_t^E \right]. \tag{2.1}$$

Although this formulation is standard, learning becomes difficult when rewards are sparse. In such cases, the agent may take many actions before receiving any meaningful signal indicating whether the trajectory is promising. Classical exploration strategies such as ϵ-greedy can provide randomization, but they do not explicitly guide the agent toward informative experiences when the state space is large or partially observable [4].

2.2 Intrinsic Motivation for Exploration

Intrinsic motivation addresses sparse rewards by supplementing the environment reward with an additional exploration bonus [5, 6]. In general, the agent receives both an extrinsic reward R_t^E and an intrinsic reward R_t^I, and optimization is performed on a combined objective:

$$\bar{r}_t = R_t^E + \beta R_t^I, \tag{2.2}$$

where $\beta \geq 0$ controls the importance of intrinsic motivation.

2.2.1 Count-Based Exploration

Count-based methods encourage exploration by assigning larger rewards to states that have been visited less frequently. In tabular settings, the intrinsic reward is often inversely proportional to the square root of the visit count. This idea has been extended to high-dimensional observations through pseudo-counts and density models [7–9]. These approaches are conceptually simple and often effective, but the quality of the reward signal depends on the ability to estimate novelty reliably in complex observation spaces.

2.2.2 Curiosity-Driven Exploration

Curiosity-driven methods define intrinsic rewards through prediction error. The Intrinsic Curiosity Module (ICM) learns a compact state representation and predicts the next latent feature from the current feature and action [10]. The prediction error of this forward model becomes an intrinsic bonus. Random Network Distillation (RND) uses the discrepancy between a fixed random target network and a learned predictor as a novelty measure [11]. RIDE rewards changes in a learned representation that are causally induced by the agent's actions [12]. A common advantage of these methods is that intrinsic rewards tend to decrease naturally as states become familiar, reducing repeated exploration of the same region.

2.3 Adaptive Weighting of Intrinsic Rewards

Despite the success of intrinsic motivation, many methods still depend on a manually chosen coefficient β [10, 11]. This fixed coefficient creates an important limitation. If β is too small, the intrinsic reward becomes negligible and exploration remains weak. If β is too large, curiosity may dominate the training signal and distract the agent from the task objective. Since the optimal trade-off differs across environments and over the course of training, fixed weighting lacks flexibility.

Several studies have attempted to overcome this issue. Extrinsic-Intrinsic Policy Optimization (EIPO) treats the trade-off as a constrained optimization problem and adapts the influence of intrinsic rewards at a global level [13]. Automatic Intrinsic Reward Shaping (AIRS) frames the selection among different intrinsic reward functions as a bandit-style decision process [14]. Population-based approaches such as Never Give Up and Agent57 coordinate multiple policies with different exploration strengths and discount factors [15, 16]. These methods improve adaptability, but their modulation is usually global, policy-level, or stage-level rather than state-dependent.

An important implication for this thesis is that the fixed-coefficient problem is shared across module families. Count-based bonuses, ICM, and RIDE differ in how they define novelty or usefulness, but they are still typically injected into the RL objective through the same scalar mixing pattern. This creates a natural experimental perspective: if ACWI is truly module-agnostic, its value should be understood in terms of how it changes the behavior of different intrinsic reward generators rather than only how it improves one specific curiosity architecture.

2.4 Research Gap

The literature suggests that intrinsic motivation is effective when the exploration bonus is aligned with future task success. However, existing methods often lack a direct mechanism for deciding *where* intrinsic rewards should be emphasized. Two states can have similar novelty scores while contributing very differently to downstream extrinsic return. A uniform or globally adapted coefficient cannot express this difference.

The central gap addressed in this thesis is therefore twofold. First, there is a lack of lightweight, state-dependent mechanisms that modulate intrinsic rewards according to how well they predict future extrinsic outcomes. Second, there is limited experimental framing for understanding whether the same adaptive mechanism can improve multiple intrinsic reward modules in a consistent way. The ACWI framework fills the first gap by learning a scaling function $\beta(s_t)$ and optimizing it using a correlation-based objective. This thesis uses that framework to address the second gap by analyzing ACWI as a common adaptive layer for representative modules such as count-based bonuses, ICM, and RIDE.

2.5 Chapter Summary

This chapter reviewed the foundations required for the proposed work. Reinforcement learning in sparse-reward environments is challenging because standard exploration methods provide weak guidance. Intrinsic motivation methods such as count-based bonuses, ICM, RND, and RIDE improve exploration but usually depend on a fixed weighting coefficient. Prior adaptive methods alleviate this problem only partially because they operate at a coarse granularity. These observations motivate the methodology presented in the next chapter, where ACWI is formulated as a state-dependent scaling layer and framed as a mechanism whose effect can be studied across representative intrinsic reward modules.

Chapter 3

Methodology

3.1 Problem Formulation

This thesis considers episodic reinforcement learning with both extrinsic and intrinsic rewards. At time step t, the agent observes state s_t, samples action $a_t \sim \pi_\theta(\cdot \mid s_t)$, receives extrinsic reward R_t^E, and transitions to the next state s_{t+1}. Let $R_t^{I,m}$ denote the intrinsic reward produced by intrinsic reward module m, where m may correspond to a count-based bonus, ICM, or RIDE. Instead of using a constant coefficient, the proposed method introduces a state-dependent scaling function $\beta_\psi(s_t)$ so that the shaped reward becomes

$$\bar{r}_t = R_t^E + \alpha \beta_\psi(s_t) R_t^{I,m}, \tag{3.1}$$

where $\alpha > 0$ is a global intrinsic-strength hyperparameter and $\beta_\psi(s_t)$ is learned online.

The intuition is straightforward. If exploration from state s_t tends to lead to higher future extrinsic return, the method should increase the influence of intrinsic reward in that state. If exploration is unhelpful or noisy, the influence should be reduced. ACWI operationalizes this idea through a lightweight Beta Network trained with a correlation-based objective. This formulation deliberately separates the adaptive scaling mechanism from the underlying intrinsic reward generator so that the same ACWI layer can be studied across different module families.

3.2 Base Reinforcement Learning Algorithm

The policy optimization backbone used in this thesis is Proximal Policy Optimization (PPO) [17]. PPO is widely adopted because it offers stable on-policy optimization with relatively simple implementation. It updates the policy by maximizing a clipped surrogate objective:

$$L^{\text{PPO}}(\theta) = \mathbb{E}_t \left[\min \left(\rho_t(\theta) \hat{A}_t, \text{clip}(\rho_t(\theta), 1-\epsilon, 1+\epsilon) \hat{A}_t \right) \right], \tag{3.2}$$

where $\rho_t(\theta) = \frac{\pi_\theta(a_t|s_t)}{\pi_{\theta_{\text{old}}}(a_t|s_t)}$ is the importance ratio and $\hat{A}_t$ is the advantage estimate. The advantages are computed with Generalized Advantage Estimation (GAE) [18] using the shaped reward in Equation 3.1.

3.3 Representative Intrinsic Reward Modules

To make the experimental framing explicit, this thesis treats ACWI as a wrapper around representative intrinsic reward modules rather than as an ICM-specific add-on. The discussion centers on three module families:

- **Count-based exploration**: intrinsic reward is derived from state novelty, typically using inverse visit frequency or pseudo-counts.
- **Intrinsic Curiosity Module (ICM)**: intrinsic reward is derived from forward prediction error in a learned latent space.
- **Rewarding Impact-Driven Exploration (RIDE)**: intrinsic reward emphasizes transitions that produce meaningful changes in the learned state representation through the agent's actions.

These modules generate different exploration signals, but all of them can be combined with ACWI through the same shaped reward in Equation 3.1. Among them, ICM is used as the concrete implementation in this thesis because it provides a clean and computationally efficient test case for quantitative evaluation.

3.3.1 ICM Instantiation

To generate intrinsic rewards in the implemented system, the thesis adopts the Intrinsic Curiosity Module (ICM) [10]. ICM first encodes each state into a latent representation $\phi(s_t) \in \mathbb{R}^d$. It then uses two auxiliary models:

- a forward dynamics model f_η^F that predicts the next latent feature from the current feature and action;
- an inverse dynamics model f_η^I that predicts the action from two consecutive latent states.

The ICM loss is a weighted combination of the forward and inverse losses:

$$L_{\text{ICM}}(\eta) = \alpha_F \mathbb{E}\left[\frac{1}{2}\left\|f_\eta^F(\phi_t, a_t) - \phi_{t+1}\right\|_2^2\right] + \alpha_I \mathbb{E}\left[-\log p_\eta(a_t \mid \phi_t, \phi_{t+1})\right]. \tag{3.3}$$

The raw intrinsic reward is defined as the forward prediction error:

$$I_t = \frac{1}{2}\left\|f_\eta^F(\phi_t, a_t) - \phi_{t+1}\right\|_2^2. \tag{3.4}$$

To improve stability, the reward is standardized within each minibatch and then rectified:

$$I_t^+ = \max\left(0, \frac{I_t - \mu_I}{\sigma_I + \varepsilon}\right), \tag{3.5}$$

where μ_I and σ_I are the minibatch mean and standard deviation. This preprocessing suppresses transitions whose prediction errors are below the batch average and retains relatively surprising transitions.

3.4 Adaptive Correlation-Weighted Intrinsic Scaling

3.4.1 Beta Network

The core contribution of the thesis is a Beta Network $\beta_\psi : \mathcal{S} \to \mathbb{R}_+$ that predicts a positive intrinsic reward multiplier from the current state. The network is intentionally lightweight to keep the computational overhead low. In the experiments adapted from the paper, the network uses an encoder depth of 2, an embedding size of 256, and clips its output to the interval $[0.1, 2.0]$.

For a chosen intrinsic reward module m, the scaled intrinsic reward is

$$\tilde{I}_t^m = \beta_\psi(s_t) I_t^{+,m}, \tag{3.6}$$

and the final reward used by PPO is

$$\bar{r}_t = R_t^E + \alpha \tilde{I}_t^m. \tag{3.7}$$

This notation emphasizes that ACWI does not define novelty by itself. Instead, it adaptively regulates the contribution of whichever intrinsic module supplies the raw bonus. In the present implementation, $I_t^{+,m}$ corresponds to the standardized ICM reward.

3.4.2 Correlation-Based Objective

Training β_ψ with meta-gradients would be computationally expensive. Instead, ACWI introduces a first-order objective that directly aligns the scaled intrinsic reward with discounted future extrinsic return. Let

$$G_t^E = \sum_{k=t}^{T} \gamma^{k-t} R_k^E \tag{3.8}$$

denote the discounted extrinsic return from time step t onward.

Within a minibatch, both the scaled intrinsic reward and the extrinsic return are standardized:

$$\hat{I}_t^m = \frac{\tilde{I}_t^m - \mu_{\tilde{I}^m}}{\sigma_{\tilde{I}^m}}, \qquad \hat{G}_t = \frac{G_t^E - \mu_G}{\sigma_G}. \tag{3.9}$$

The correlation loss is then defined as

$$L_{\mathrm{corr}}(\psi) = -\mathbb{E}[\hat{I}_t^m \hat{G}_t]. \tag{3.10}$$

Minimizing this loss encourages $\beta_\psi(s_t)$ to be larger in states where intrinsic exploration is positively associated with future extrinsic success.

To avoid unstable or degenerate scaling, ACWI also uses a regularization term in log-space:

$$L_{\mathrm{reg}}(\psi) = \mathbb{E}\left[\left(\log \beta_\psi(s_t) - \log \beta_0\right)^2\right], \tag{3.11}$$

where $\beta_0 = 1.0$ is the reference value. The complete objective is

$$L_\beta(\psi) = L_{\mathrm{corr}}(\psi) + \lambda_{\mathrm{reg}} L_{\mathrm{reg}}(\psi). \tag{3.12}$$

3.5 Training Procedure

The complete training loop integrates PPO, an intrinsic reward module, and ACWI as follows:

1. collect on-policy trajectories using the current policy;
2. update the chosen intrinsic reward module on the collected batch to compute intrinsic rewards;
3. standardize and rectify the intrinsic reward signal for that module;

4. compute discounted extrinsic returns for each time step;
5. update the Beta Network by minimizing the correlation-based objective;
6. construct the shaped reward with the learned state-dependent scaling;
7. compute GAE advantages from the shaped reward;
8. update the policy and value networks with PPO for several epochs.

This procedure preserves compatibility with conventional PPO training while adding only a small auxiliary network and a simple optimization step before each PPO update cycle. In the implemented experiments, the intrinsic module in Step 2 is ICM.

3.6 Discussion

The methodology offers three important advantages. First, it avoids manual selection of a single intrinsic reward coefficient for all states. Second, it remains computationally efficient because it does not require unrolled policy optimization or high-order differentiation. Third, it is modular: the scaling mechanism is agnostic to the base RL algorithm and can be paired with other intrinsic reward generators besides ICM. This modularity is central to the thesis framing because it allows ACWI to be analyzed as a common adaptive layer for count-based, ICM, and RIDE-style exploration signals.

At the same time, the method assumes that at least some correlation exists between local exploration and future extrinsic return. When the environment is so sparse that extrinsic signals remain near zero for long periods, the Beta Network receives little informative supervision. In such cases, the method is expected to behave similarly to a fixed scaling strategy rather than producing strong adaptation. This limitation becomes important in the empirical analysis of Chapter 4 and helps explain why the impact of ACWI may differ across intrinsic reward modules.

Chapter 4

Experimental Setup and Results

4.1 Benchmark Environments

The empirical evaluation is conducted on five sparse-reward MiniGrid environments [19]. These environments are chosen because they represent different forms of exploration difficulty while maintaining a common gridworld interface.

Bảng 4.1: MiniGrid environments used in the experiments

Environment	**Purpose in the evaluation**
DoorKey-8x8	Tests whether the agent can discover a key, unlock a door, and shift from exploration to goal-directed behavior.
Empty-16x16	Represents an extreme sparse-reward setting with almost no intermediate feedback before the goal is reached.
RedBlueDoors-8x8	Requires opening two doors in the correct order, testing whether the method can suppress irrelevant exploration.
UnlockPickup	Extends DoorKey with an additional object manipulation stage, creating a hierarchical sparse-reward task.
KeyCorridorS3R3	A longer-horizon task with multiple rooms, keys, and locked doors, used to assess sustained exploration ability.

These tasks share several useful properties: sparse terminal rewards, egocentric partial observations, step penalties that discourage random motion, and stochastic initialization over multiple random seeds. Together, they form a suitable test bed for evaluating whether adaptive intrinsic reward scaling improves both efficiency and robustness.

4.2 Baselines and Evaluation Criteria

The experimental framing of this thesis treats ACWI as an adaptive layer that can be placed on top of representative intrinsic reward modules. The three module families emphasized throughout the thesis are count-based bonuses, ICM, and RIDE. In the present implementation, the reported quantitative experiments instantiate the ICM branch of this framing, because it provides a controlled and computationally manageable case for measuring the effect of adaptive scaling.

Under that implementation, the proposed ACWI method is compared against two categories of baselines:

- PPO trained only with extrinsic rewards;
- PPO combined with ICM using fixed intrinsic coefficients $\beta \in \{0.1, 0.2, 0.5, 1, 2\}$.

This comparison is important because it isolates the contribution of adaptive scaling from the contribution of intrinsic motivation itself. PPO provides a lower bound on exploration ability, while fixed-coefficient PPO+ICM shows how sensitive performance is to manual tuning. From the perspective of the broader thesis framing, the comparison also serves as the primary case study for understanding how ACWI affects an intrinsic reward module once the module's raw signal is held fixed.

The evaluation focuses on four aspects:

- episode return over training time;
- sample efficiency in the early and middle stages of learning;
- variance across random seeds;
- qualitative behavior of the learned scaling coefficient $\beta(s)$ and what it reveals about module-dependent exploration control.

4.3 Experimental Configuration

All reported runs share the same PPO backbone and ICM architecture so that the comparison is fair. The adapted paper fixes the global intrinsic coefficient at $\alpha = 0.001$ and clips the learned Beta Network output to $[0.1, 2.0]$. The Beta Network uses a hidden representation size of 256 with encoder depth 2. The regularization weight for the correlation objective is $\lambda_{\text{reg}} = 10^{-3}$, the reference value is $\beta_0 = 1.0$, gradient clipping is set to 1.0, and weight decay is 10^{-6}.

Although the current implementation focuses on ICM, the same ACWI formulation would be applied to count-based and RIDE modules by replacing the raw intrinsic reward source while keeping the state-dependent scaling and correlation objective unchanged. This is why the chapter is written not only as a report of one implementation, but also as an empirical framing for studying ACWI across intrinsic reward modules.

Bảng 4.2: Key hyperparameters reported for ACWI

Hyperparameter	**Value**
Intrinsic strength α	0.001
Beta output bounds	$[0.1, 2.0]$
Encoder size	256
Encoder depth	2
Regularization weight λ_{reg}	10^{-3}
Reference value β_0	1.0
Gradient clipping	1.0
Weight decay	10^{-6}

The results are reported over five random seeds. This setup is sufficient to assess whether adaptive scaling consistently improves learning behavior rather than benefiting from a single favorable initialization.

4.4 Results

The quantitative experiments reported in this thesis correspond to the ICM instantiation of ACWI. Within that setting, the results show several consistent trends. First, ACWI performs strongly across environments with sparse but still informative extrinsic rewards. In DoorKey-8x8, RedBlueDoors-8x8, UnlockPickup, and KeyCorridorS3R3, the adaptive approach achieves faster learning and lower variance than most fixed-coefficient baselines. This result indicates that ACWI improves sample efficiency and makes exploration behavior more stable across runs.

Second, the comparison against fixed-coefficient PPO+ICM highlights the difficulty of manual tuning. A coefficient that works well in one environment can be too small or too large in another. Small values underemphasize exploration and slow down learning, while large values can inject excessive noise and interfere with exploitation. ACWI mitigates both problems by learning to amplify intrinsic rewards only in states where they correlate with future extrinsic success. In the broader module-level framing of this thesis, this result supports the hypothesis that ACWI is most useful when an intrinsic reward signal is informative but unevenly relevant across the state space.

Third, the method exhibits a natural transition from exploration to exploitation. As training progresses and the policy begins to discover task-relevant behaviors, the effective intrinsic contribution tends to decrease. This suggests that ACWI does not merely add curiosity-driven noise, but instead learns to reduce exploration pressure once the task structure becomes clearer.

4.5 Analysis of the Learned Scaling Behavior

The paper reports that the learned distribution of $\beta(s)$ evolves differently across environments. In DoorKey-8x8 and RedBlueDoors-8x8, the distribution is initially concentrated near its reference value, then gradually broadens and becomes more structured. This pattern suggests that the Beta Network learns to partition the state space into regions where different exploration intensities are appropriate. Toward the end of training, the distribution shifts toward lower values, reflecting the decreasing marginal value of intrinsic rewards once the policy has matured.

In contrast, Empty-16x16 behaves differently. Because extrinsic rewards remain almost zero for most trajectories, the correlation objective receives very weak supervision. As a result, the Beta Network stays close to its initialization and exhibits limited adaptation. Importantly, this failure mode is graceful rather than unstable: the method effectively defaults to a nearly fixed intrinsic coefficient instead of diverging. This observation confirms both a strength and a limitation of ACWI. The method is robust when correlation signals are weak, but meaningful adaptation requires at least occasional extrinsic feedback. From a module-level perspective, this also suggests that ACWI will be most effective for intrinsic reward modules whose signals can be related to eventual task progress often enough for the correlation objective to learn from experience.

4.6 Discussion

Overall, the empirical evidence supports the main claim of the thesis: state-dependent intrinsic reward scaling can improve sparse-reward reinforcement learning when the environment provides enough structure for the agent to relate exploration to eventual success. In the implemented ICM case study, ACWI is especially beneficial in environments where exploration needs change across states and over time. The method is less effective in extremely sparse tasks where the extrinsic return is too weak to supervise adaptation. Therefore, ACWI should be viewed as a practical middle ground between fixed intrinsic scaling and more expensive meta-learning approaches, and as a promising adaptive wrapper for broader module families such as count-based bonuses and RIDE.

Conclusion and Future Work

This thesis studies Adaptive Correlation-Weighted Intrinsic Rewards (ACWI) in reinforcement learning. Instead of treating ACWI as a method used with only one type of curiosity reward, this work views it as an adaptive layer to study how state-dependent scaling of intrinsic rewards affects learning. The thesis considers three main types of intrinsic rewards: count-based bonuses, ICM, and RIDE. ACWI is used as the mechanism that decides how much these rewards should influence learning in each state. By matching the scaled intrinsic reward with the discounted future extrinsic return, ACWI offers a simple way to balance exploration and exploitation during training without using costly meta-gradient methods.

Experiments on MiniGrid use PPO+ICM as the main case study and show that ACWI improves learning stability and sample efficiency in several sparse-reward tasks, especially when extrinsic rewards are rare but still useful. Compared to fixed coefficients that need manual tuning, the adaptive approach works more consistently across different environments and random seeds. However, the results also show a limitation: when extrinsic rewards are almost always missing, the correlation signal becomes weak, and the scaling factor behaves like a fixed coefficient. This suggests that the effectiveness of ACWI depends on how useful the intrinsic reward is and how often it relates to future task success.

There are several directions for future work. First, the experiments can be extended by testing ACWI more fully on count-based bonuses and RIDE, not only ICM. Second, the method can be applied to broader settings such as continuous control, multi-task RL, and embodied navigation. Third, a clearer theoretical analysis of the correlation objective could help explain when adaptive scaling is most useful across different intrinsic reward types. These directions can further improve the use of adaptive intrinsic rewards in reinforcement learning.

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